

QUANTITATIVE FINANCE

COMPLETE THEORY REFERENCE

Every Mathematical Theory Used by
Professional Quants for Real-Time Prediction

Renaissance Technologies | Citadel | DE Shaw | Two Sigma | Jane Street

Part 1: Core Theories (25 Categories, 150+ Methods)

Part 2: Advanced & Emerging Theories

Summary Table: Who Uses What

PART 1: CORE THEORIES

1. STOCHASTIC CALCULUS & DIFFUSION PROCESSES

* Geometric Brownian Motion (GBM)

$dS = \mu S dt + \sigma S dW$. Log-normal prices. Foundation of all pricing.

* Ito's Lemma

Stochastic chain rule with extra variance term. Derives Black-Scholes.

* Stochastic Differential Equations (SDEs)

General framework for random price evolution.

* Merton Jump-Diffusion (1976)

GBM + Poisson jumps for sudden crashes and gaps.

* Kou Double-Exponential Jump-Diffusion

Asymmetric jumps. Downward jumps larger than upward.

* Levy Processes

Variance Gamma, Normal Inverse Gaussian, CGMY. Generalized jumps.

* Fractional Brownian Motion (fBM)

Long memory. Hurst $H > 0.5$ = trending, $H < 0.5$ = mean-reverting.

* Ornstein-Uhlenbeck Process

Mean-reverting. Used in pairs trading, interest rate modeling.

2. STOCHASTIC VOLATILITY MODELS

* Heston Model (1993)

Vol is random: $dv = k(\theta - v)dt + \xi \sqrt{v}dW$. Industry standard.

* SABR Model

Stochastic Alpha Beta Rho. Standard for IR and FX options.

* Local Volatility (Dupire)

$\text{Vol} = f(S, t)$. Fits all market option prices exactly.

* CEV Model

Constant Elasticity of Variance. Vol depends on price level.

* Rough Volatility (2014+)

Fractional process $H \sim 0.1$. Cutting-edge. Better VIX prediction.

* 3/2 Stochastic Volatility Model

dv proportional to $v^{3/2}$. Better tail behavior.

3. GARCH FAMILY (VOLATILITY FORECASTING)

* GARCH(1,1)

$\sigma^2_t = \omega + \alpha_1 \epsilon^2_{t-1} + \beta_1 \sigma^2_{t-1}$. Workhorse model.

* EGARCH

Exponential GARCH. Captures leverage effect (drops = higher vol).

* GJR-GARCH

Threshold GARCH. Asymmetric shock impact.

* FIGARCH

Fractionally Integrated. Long memory in volatility.

* TGARCH

Threshold variant. Different coefficients for pos/neg shocks.

* DCC-GARCH

Dynamic Conditional Correlation. Time-varying cross-asset correlations.

* BEKK-GARCH

Multivariate. Covariance matrix forecasting.

* Realized GARCH

Uses high-frequency realized vol as input.

4. MONTE CARLO METHODS

* **Standard Monte Carlo**

Simulate 10,000+ price paths. Count outcomes for probability.

* **Quasi-Monte Carlo (Sobol/Halton)**

Low-discrepancy sequences. Faster convergence.

* **Importance Sampling**

Oversample rare tail events. Critical for risk estimation.

* **Antithetic Variates**

Negatively correlated pairs reduce variance.

* **Control Variates**

Use known quantities to reduce estimation error.

* **Stratified Sampling**

Divide probability space into strata.

* **Least-Squares MC (Longstaff-Schwartz)**

American option pricing via regression.

* **MCMC (Markov Chain Monte Carlo)**

Metropolis-Hastings, Gibbs. Sample complex posteriors.

* **Particle Filtering (Sequential MC)**

Real-time non-linear state estimation.

5. BAYESIAN METHODS

- * **Bayes' Theorem**
 $P(H|D) = P(D|H) \cdot P(H) / P(D)$. Update beliefs with evidence.
- * **Bayesian Linear Regression**
Posterior over parameters. Quantifies uncertainty.
- * **Bayesian Model Averaging**
Average all models weighted by evidence.
- * **Bayesian Networks**
DAGs modeling causal economic relationships.
- * **Thompson Sampling**
Exploration vs exploitation for strategy selection.
- * **Bayesian Signal Fusion**
Combine multiple agent signals with proper uncertainty.

6. STATE ESTIMATION & FILTERING

- * **Kalman Filter**
Optimal linear state estimation from noisy data.
- * **Extended Kalman Filter (EKF)**
Non-linear via Jacobian linearization.
- * **Unscented Kalman Filter (UKF)**
Sigma points for better non-linear handling.
- * **Hidden Markov Models (HMM)**
Detect hidden regimes: Bull, Bear, Crisis.
- * **Forward-Backward Algorithm**
State probabilities given all observations.
- * **Viterbi Algorithm**
Most likely sequence of hidden states.
- * **Baum-Welch (EM) Algorithm**
Learn HMM parameters from data.

7. TIME SERIES MODELS

- * **ARIMA / SARIMA**
Autoregressive + Moving Average. Trend + seasonality.
- * **VAR (Vector Autoregression)**
Multi-variable. Cross-asset interactions.
- * **VECM (Vector Error Correction)**
VAR with cointegration constraints.
- * **State-Space Models**
General: Kalman filter + flexible dynamics.
- * **Regime-Switching (Hamilton)**
Markov-switching parameters.
- * **Threshold Autoregression (TAR)**
Different dynamics above/below threshold.
- * **ARFIMA (Long Memory)**
Fractional integration for persistent series.

8. OPTIONS & DERIVATIVES PRICING

- * **Black-Scholes-Merton Formula**
 $C = S \cdot N(d1) - K \cdot \exp(-rT) \cdot N(d2)$.
- * **Binomial Tree (Cox-Ross-Rubinstein)**

Discrete-time. Converges to Black-Scholes.

*** Risk-Neutral Pricing**

Price = E^Q [discounted payoff]. Fundamental theorem.

*** The Greeks**

Delta, Gamma, Theta, Vega, Rho. Option sensitivities.

*** Implied Volatility Surface**

Vol varies by strike/expiry. Smile, skew.

*** Breeden-Litzenberger Formula**

Extract full probability density from option prices.

*** Variance Swaps & Vol Derivatives**

Trade volatility directly. VIX methodology.

9. FACTOR MODELS & ALPHA GENERATION

* CAPM

$E[R] = R_f + \beta(R_m - R_f)$. Single factor.

* Fama-French 3-Factor

Market + Size (SMB) + Value (HML).

* Fama-French 5-Factor

+ Profitability (RMW) + Investment (CMA).

* Carhart 4-Factor

FF3 + Momentum (MOM).

* Arbitrage Pricing Theory (APT)

Generalized multi-factor framework.

* Barra Risk Model

Industry standard. 70+ factors.

* PCA (Principal Component Analysis)

Extract latent factors from covariance matrix.

* Statistical / Eigenportfolios

Trade eigenvectors of correlation matrix.

10. PORTFOLIO OPTIMIZATION

* Mean-Variance (Markowitz)

Efficient frontier. Max return for given risk.

* Black-Litterman Model

Market equilibrium + subjective views.

* Risk Parity

Equal risk contribution from each asset.

* Kelly Criterion

Optimal bet size: $f = (bp - q)/b$. Max log wealth.

* Robust Optimization

Handle parameter uncertainty.

* LASSO / Elastic Net

Sparse portfolio selection. Regularization.

* Ledoit-Wolf Shrinkage

Better covariance estimation.

* Hierarchical Risk Parity (HRP)

Cluster-based. No matrix inversion needed.

* CVaR Optimization

Minimize conditional value-at-risk.

11. RISK MANAGEMENT

* Value at Risk (VaR)

95/99%: max loss won't exceed \$X.

* Conditional VaR / Expected Shortfall

Average loss in worst 5% scenarios.

* Extreme Value Theory (EVT)

GEV + Generalized Pareto for tails.

* Peaks Over Threshold (POT)

Model exceedances above high threshold.

* Copula Theory

Gaussian, Clayton, t-Copula. Full dependency structure.

* Stress Testing

Scenario analysis for extreme conditions.

* Drawdown Analysis

12. MARKET MICROSTRUCTURE

- * **Kyle's Lambda**

Price impact per unit of order flow.

- * **Amihud Illiquidity Ratio**

Price impact per dollar of volume.

- * **Roll's Spread Estimator**

Estimate bid-ask from transaction data.

- * **Limit Order Book (LOB) Models**

Buy/sell order dynamics.

- * **Hawkes Processes**

Self-exciting events. Trade clustering.

- * **Almgren-Chriss Optimal Execution**

Min market impact for large orders.

- * **VWAP / TWAP**

Volume/Time weighted execution algos.

- * **Avellaneda-Stoikov Market Making**

Optimal bid-ask quoting.

13. STATISTICAL ARBITRAGE

- * **Cointegration (Engle-Granger)**
Stationary spread. Pairs trading foundation.
- * **Johansen Test**
Multi-variable cointegration vectors.
- * **Ornstein-Uhlenbeck for Spreads**
Mean-reverting spread. Half-life estimation.
- * **Distance Method**
Find pairs by minimizing squared price differences.
- * **Copula-Based Pairs Trading**
Dependency modeling for mispricing detection.

14. MACHINE LEARNING FOR FINANCE

- * **Logistic Regression**
Up/down probability classification.
- * **Random Forest / XGBoost / LightGBM**
Gradient boosting. Feature importance.
- * **LSTM (Long Short-Term Memory)**
RNN for sequential price data.
- * **Temporal Transformers**
Attention for time series. State-of-the-art.
- * **Autoencoders**
Anomaly detection, dimensionality reduction.
- * **Graph Neural Networks (GNN)**
Stock relationships as graph.
- * **Gaussian Processes**
Non-parametric Bayesian regression with uncertainty.

15. REINFORCEMENT LEARNING FOR TRADING

- * **DQN (Deep Q-Network)**
Discrete: buy/sell/hold.
- * **PPO (Proximal Policy Optimization)**
Continuous position sizing.
- * **SAC (Soft Actor-Critic)**
Entropy-regularized exploration.
- * **A2C / A3C**
Actor-Critic parallel training.
- * **Multi-Agent RL**
Multiple agents, shared information.
- * **Reward Shaping**
 $\text{PnL} - \text{risk} - \text{costs} + \text{Sharpe contribution}$.
- * **Inverse RL**
Learn reward from expert trader behavior.

16. INFORMATION THEORY

- * **Shannon Entropy**
Market uncertainty/disorder measure.
- * **Transfer Entropy**
Directional causal flow. News $\rightarrow$ Price.
- * **Mutual Information**

Non-linear dependence. Beyond correlation.

* **KL Divergence**

Distance between distributions. Regime change.

* **Maximum Entropy Principle**

Least-assuming distribution when uncertain.

17. CAUSAL INFERENCE

* **Granger Causality**

Does X predict Y beyond Y's own past?

* **Do-Calculus (Pearl)**

True causal effects. $P(Y|do(X))$.

* **Instrumental Variables**

Handle confounders in causal analysis.

* **Difference-in-Differences**

Event study: policy impact on prices.

* **Synthetic Control**

Counterfactual what-if analysis.

* **Propensity Score Matching**

Match for causal comparison.

18. SIGNAL PROCESSING

- * **Fast Fourier Transform (FFT)**
Frequency decomposition. Detect hidden cycles.
- * **Wavelet Transform**
Multi-scale: daily/weekly/monthly trends simultaneously.
- * **Hilbert-Huang Transform**
Non-linear, non-stationary decomposition.
- * **Empirical Mode Decomposition**
Data-driven intrinsic mode functions.
- * **Savitzky-Golay Filter**
Polynomial smoothing. Preserves peaks.
- * **Butterworth / Chebyshev Filters**
Low/high/band-pass digital filters.

19. MATRIX & SPECTRAL METHODS

- * **Random Matrix Theory (RMT)**
Marchenko-Pastur. Signal vs noise in correlations.
- * **Spectral Analysis**
Eigenvalue decomposition of correlation matrices.
- * **Minimum Spanning Tree**
Graph of strongest correlations.
- * **Shrinkage Estimators (Ledoit-Wolf)**
Better covariance for high-dim data.
- * **Matrix Completion**
Fill missing cross-asset data.
- * **Tensor Decomposition**
Multi-way higher-order relationships.

20. STOCHASTIC OPTIMAL CONTROL

- * **Hamilton-Jacobi-Bellman (HJB)**
Master PDE for optimal sequential decisions.
- * **Pontryagin Maximum Principle**
Necessary optimality conditions.
- * **Merton's Problem**
Optimal consumption + investment over lifetime.
- * **Dynamic Programming (Bellman)**
Backwards-in-time sequential optimization.
- * **Model Predictive Control (MPC)**
Rolling optimization horizon.
- * **Linear-Quadratic Regulator (LQR)**
Closed-form Riccati solution.

21. GAME THEORY

- * **Nash Equilibrium**
Multiple traders optimizing simultaneously.
- * **Stackelberg Games**
Leader-follower in dark pools.
- * **Auction Theory**
Price discovery. IPOs, exchanges.
- * **Mechanism Design**

Exchange rule effects on behavior.

- * **Mean-Field Games**

Large population macro from micro.

- * **Bayesian Games**

Incomplete info. Kyle's informed trading.

22. FRACTAL & CHAOS THEORY

- * **Mandelbrot Multifractal Model (MMAR)**

Markets are fractal, not Gaussian.

- * **MF-DFA**

Multifractal Detrended Fluctuation Analysis.

- * **Lyapunov Exponents**

Chaos sensitivity. Positive = unpredictable.

- * **Hurst Exponent (R/S)**

Long-range dependence detection.

- * **Recurrence Quantification**

Recurring patterns in nonlinear dynamics.

23. AGENT-BASED MODELING

- * **Zero-Intelligence Agents**

Random traders make realistic order books.

- * **Heterogeneous Agent Models**

Fundamentalists vs chartists vs noise.

- * **Santa Fe Artificial Stock Market**

Emergent bubbles and crashes.

- * **Contagion & Cascade Models**

How crashes spread through networks.

24. TOPOLOGICAL DATA ANALYSIS (TDA)

- * **Persistent Homology**

Shape features persisting across scales.

- * **Betti Numbers**

Topological invariants for regime changes.

- * **Vietoris-Rips Complex**

Simplicial complex from market data.

- * **Mapper Algorithm**

Visualize high-dim financial data.

- * **Crash Topology**

Topology changes before crashes. JP Morgan research.

25. CUTTING EDGE & EMERGING

- * **Quantum Monte Carlo**

Quadratic speedup for option pricing.

- * **Quantum Annealing (D-Wave)**

Portfolio optimization on quantum hardware.

- * **Neural SDEs**

Neural networks + stochastic differential equations.

- * **Conformal Prediction**

Distribution-free prediction intervals.

- * **Foundation Models (FinGPT)**

Pre-trained LLMs for financial tasks.

- * **GANs for Synthetic Data**

TimeGAN: realistic market scenarios.

- * **Normalizing Flows**

Exact density estimation. Complex distributions.

- * **Graph Attention Networks**

Attention-weighted inter-stock relationships.

PART 2: ADVANCED THEORIES

26. MARTINGALE THEORY & MEASURE THEORY

- * **Risk-Neutral Measure (Q-Measure)**

Price = $E^Q[\text{payoff}]$. Not real-world P. Foundation of pricing.

- * **Girsanov's Theorem**

Switch between P and Q measures. Enables risk-neutral pricing.

- * **Martingale Representation Theorem**

Any claim replicable by self-financing strategy.

- * **Radon-Nikodym Derivative (dQ/dP)**

Mathematical bridge between probability measures.

- * **Fundamental Theorem of Asset Pricing**

No-arbitrage = existence of equivalent martingale measure.

- * **Optional Stopping Theorem**

$E[M_\tau] = E[M_0]$. Used in barrier options.

27. PDE METHODS

- * **Black-Scholes PDE**

$dV/dt + 0.5\sigma^2 S^2 d^2V/dS^2 + rS dV/dS - rV = 0$

- * **Feynman-Kac Formula**

Links PDEs to stochastic expectations.

- * **Finite Difference Methods**

Explicit, implicit, Crank-Nicolson numerical schemes.

- * **Free Boundary Problems**

American options. Exercise boundary part of solution.

- * **Viscosity Solutions**

Weak PDE solutions for HJB equations.

28. RANDOM MATRIX THEORY (RMT)

- * **Marchenko-Pastur Distribution**

Theoretical eigenvalue dist. Signal vs noise.

- * **Noise Filtering**

Remove spurious correlations. Used at RenTech.

- * **Wishart Distribution**

Distribution of sample covariance matrices.

- * **Tracy-Widom Distribution**

Largest eigenvalue. Detect significant factors.

- * **Correlation Matrix Cleaning**

Replace noise eigenvalues with RMT predictions.

29. SURVIVAL ANALYSIS

- * **Kaplan-Meier Estimator**

Non-parametric survival curve. Time to target price.

- * **Cox Proportional Hazards**

Which factors accelerate time-to-event?

- * **Competing Risks**

Stop-loss vs take-profit: which comes first?

- * **Accelerated Failure Time (AFT)**

Parametric. Factors affect time multiplicatively.

30. COMPRESSED SENSING & SPARSE RECOVERY

- * **Compressed Sensing**

Reconstruct signals from limited measurements.

- * **Matrix Completion**

Fill missing cross-asset correlation data.

- * **Sparse Factor Models**

L1-penalized factor analysis.

- * **Robust PCA**

Low-rank + sparse decomposition.

31. INFORMATION GEOMETRY

- * **Fisher Information Matrix**

Curvature of statistical model space.

- * **Natural Gradient Descent**

Faster optimization on probability manifolds.

- * **Geodesic Distance**

Better than KL divergence between distributions.

- * **Exponential / Mixture Families**

Geometric structure of finance distributions.

COMPLETE SUMMARY: WHO USES WHAT

Category	Key Theories	Used By
Foundations	Martingale, Measure Theory, Ito	Everyone
Pricing	Black-Scholes PDE, Feynman-Kac, FDM	Derivatives desks
Volatility	GARCH, Heston, SABR, Rough Vol	All quants
Simulation	Monte Carlo, Particle Filter, MCMC	All quants
Statistics	Bayesian, Kalman, HMM	Systematic funds
Factors	Fama-French, PCA, Barra, APT	Long/short equity
Matrix Methods	RMT, Shrinkage, Spectral Analysis	Top quant funds
Optimization	Convex, Black-Litterman, LASSO	Portfolio managers
Control	HJB, Dynamic Programming, Merton	Execution, hedging
Microstructure	Hawkes, Kyle Lambda, LOB, Almgren	HFT firms
Tail Risk	EVT, Copulas, Multifractal	Risk management
ML / RL	DQN, PPO, Transformers, GNN	Modern quant funds
Game Theory	Nash, Stackelberg, Auction	Market makers
Signal Processing	Fourier, Wavelet, Kalman	Signal traders
Causal	Granger, Do-Calculus, Transfer Entropy	Research teams
Time Series	ARIMA, GARCH, VAR, HMM	All quants
Stat Arb	Cointegration, O-U, Pairs Trading	Stat arb funds
Survival	Kaplan-Meier, Cox, Competing Risks	Risk analysts
Cutting Edge	TDA, Quantum, Neural SDE, FinGPT	R&D labs

Total: 31 categories | 200+ individual theories and methods | Covers all quant specializations: derivatives, stat arb, HFT, risk management, portfolio construction, optimal execution, and research.